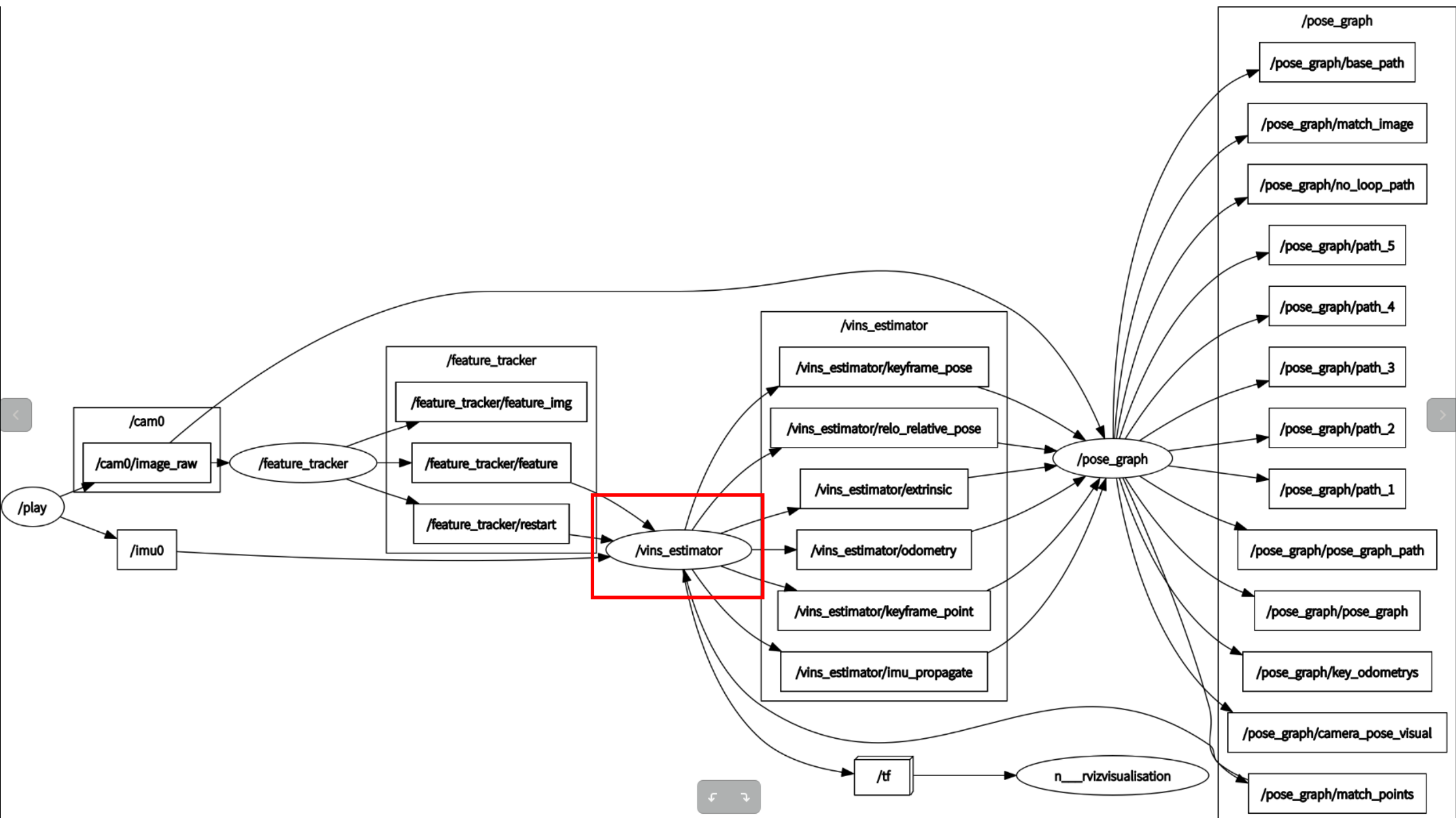
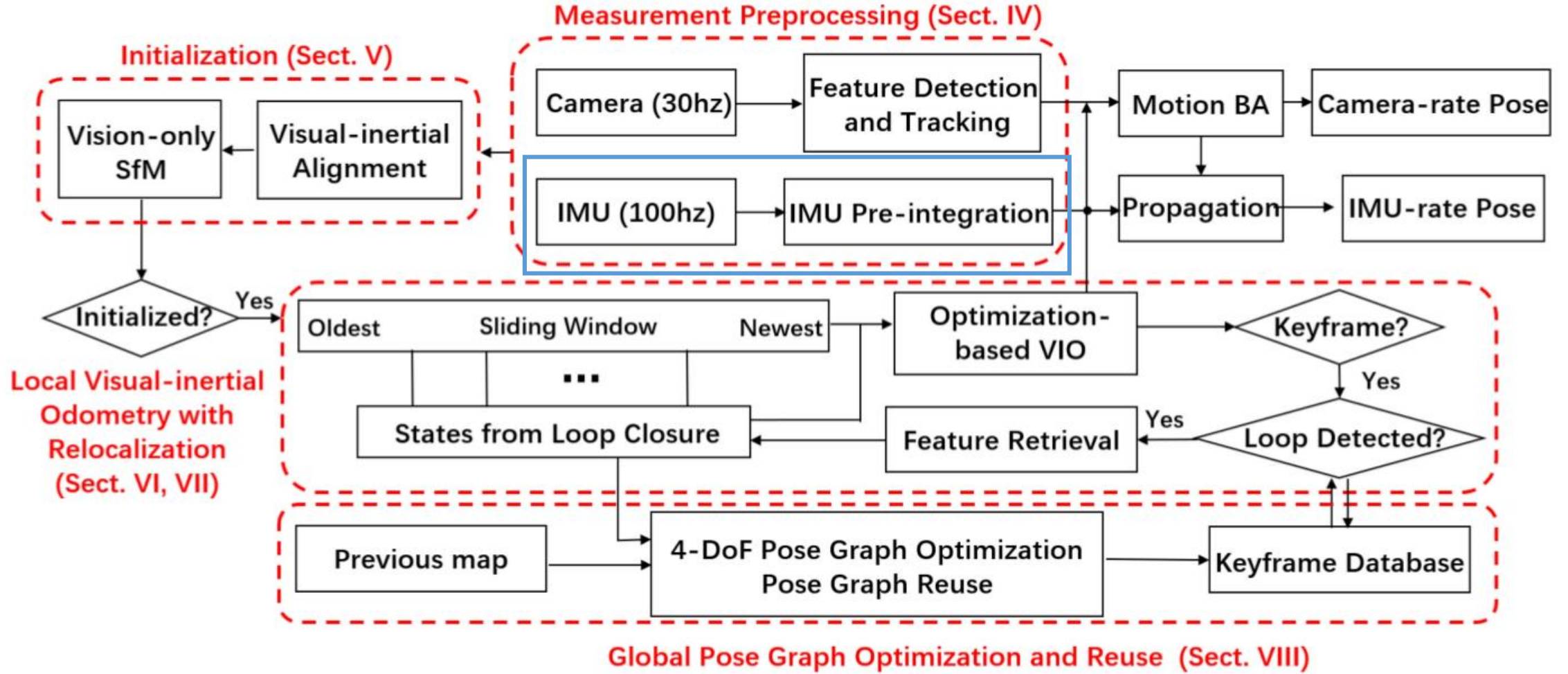


IMU预积分



课程介绍





旋转的几种数学表示方法

旋转矩阵：运算方便;使用9个量描述3自由度旋转，引入额外约束，求导困难

旋转向量：紧凑的表示旋转;具有周期性

欧拉角：直观，用户友好;需要指定旋转顺序，存在万象节死锁问题，无法进行球面平滑插值

四元数：紧凑的不带奇异的表示方法;对用户来讲不太直观，要求单位四元数，表示旋转也会过参数化

连续时间的PVQ积分

位置和速度的连续时间积分

$$\begin{aligned}\mathbf{p}_{b_k+1}^w &= \mathbf{p}_{b_k}^w + \mathbf{v}_{b_k}^w \Delta t_k \\ &\quad + \iint_{t \in [t_k, t_{k+1}]} (\mathbf{R}_t^w (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) - \mathbf{g}^w) dt^2 \\ \mathbf{v}_{b_k+1}^w &= \mathbf{v}_{b_k}^w + \int_{t \in [t_k, t_{k+1}]} (\mathbf{R}_t^w (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) - \mathbf{g}^w) dt\end{aligned}$$

四元数的乘法

$$\mathbf{p} \otimes \mathbf{q} = \begin{bmatrix} p_w q_w - p_x q_x - p_y q_y - p_z q_z \\ p_w q_x + p_x q_w + p_y q_z - p_z q_y \\ p_w q_y - p_x q_z + p_y q_w + p_z q_x \\ p_w q_z + p_x q_y - p_y q_x + p_z q_w \end{bmatrix}.$$

$$\mathbf{q}_1 \otimes \mathbf{q}_2 = [\mathbf{q}_1]_L \mathbf{q}_2 \quad \text{and} \quad \mathbf{q}_1 \otimes \mathbf{q}_2 = [\mathbf{q}_2]_R \mathbf{q}_1$$

$$[\mathbf{q}]_L = \begin{bmatrix} q_w & -q_x & -q_y & -q_z \\ q_x & q_w & -q_z & q_y \\ q_y & q_z & q_w & -q_x \\ q_z & -q_y & q_x & q_w \end{bmatrix}, \quad [\mathbf{q}]_R = \begin{bmatrix} q_w & -q_x & -q_y & -q_z \\ q_x & q_w & q_z & -q_y \\ q_y & -q_z & q_w & q_x \\ q_z & q_y & -q_x & q_w \end{bmatrix}$$

$$[\mathbf{q}]_L = q_w \mathbf{I} + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & [\mathbf{q}_v]_\times \end{bmatrix}, \quad [\mathbf{q}]_R = q_w \mathbf{I} + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -[\mathbf{q}_v]_\times \end{bmatrix}$$

$$[\mathbf{a}]_\times \triangleq \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

$$[\mathbf{a}]_\times \mathbf{b} = \mathbf{a} \times \mathbf{b} \; , \quad \forall \mathbf{a}, \mathbf{b} \in \mathbb{R}^3 \; .$$

四元数求导推导

位置和速度的连续时间积分

$$\mathbf{q}_{b_{k+1}}^w = \mathbf{q}_{b_k}^w \otimes \int_{t \in [t_k, t_{k+1}]} \frac{1}{2} \boldsymbol{\Omega}(\hat{\boldsymbol{\omega}}_t - \mathbf{b}_{w_t} - \mathbf{n}_w) \mathbf{q}_t^{b_k} dt \quad (23)$$

where

$$\boldsymbol{\Omega}(\boldsymbol{\omega}) = \begin{bmatrix} -[\boldsymbol{\omega}]_{\times} & \boldsymbol{\omega} \\ -\boldsymbol{\omega}^T & 0 \end{bmatrix}, [\boldsymbol{\omega}]_{\times} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}. \quad (24)$$

Δt_k is the duration between the time interval $[t_k, t_{k+1}]$. v

滤波问题中imu积分

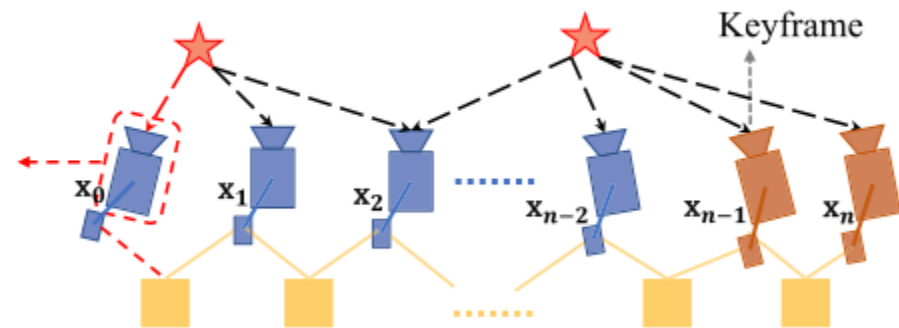
$$\begin{aligned}\mathbf{p}_{b_{k+1}}^w &= \mathbf{p}_{b_k}^w + \mathbf{v}_{b_k}^w \Delta t_k \\ &\quad + \iint_{t \in [t_k, t_{k+1}]} (\mathbf{R}_t^w (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) - \mathbf{g}^w) dt^2 \\ \mathbf{v}_{b_{k+1}}^w &= \mathbf{v}_{b_k}^w + \int_{t \in [t_k, t_{k+1}]} (\mathbf{R}_t^w (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) - \mathbf{g}^w) dt \\ \mathbf{q}_{b_{k+1}}^w &= \mathbf{q}_{b_k}^w \otimes \int_{t \in [t_k, t_{k+1}]} \frac{1}{2} \boldsymbol{\Omega}(\hat{\boldsymbol{\omega}}_t - \mathbf{b}_{w_t} - \mathbf{n}_w) \mathbf{q}_t^{b_k} dt \quad (23)\end{aligned}$$

但是优化问题中怎么办呢？

预积分的形成

$$\begin{aligned}
 \mathbf{R}_w^{b_k} \mathbf{p}_{b_{k+1}}^w &= \mathbf{R}_w^{b_k} \left(\mathbf{p}_{b_k}^w + \mathbf{v}_{b_k}^w \Delta t_k - \frac{1}{2} \mathbf{g}^w \Delta t_k^2 \right) + \boldsymbol{\alpha}_{b_{k+1}}^{b_k} \\
 \mathbf{R}_w^{b_k} \mathbf{v}_{b_{k+1}}^w &= \mathbf{R}_w^{b_k} (\mathbf{v}_{b_k}^w - \mathbf{g}^w \Delta t_k) + \boldsymbol{\beta}_{b_{k+1}}^{b_k} \\
 \mathbf{q}_w^{b_k} \otimes \mathbf{q}_{b_{k+1}}^w &= \boldsymbol{\gamma}_{b_{k+1}}^{b_k}
 \end{aligned} \tag{25}$$

$$\begin{aligned}
 \boldsymbol{\alpha}_{b_{k+1}}^{b_k} &= \iint_{t \in [t_k, t_{k+1}]} \mathbf{R}_t^{b_k} (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) dt^2 \\
 \boldsymbol{\beta}_{b_{k+1}}^{b_k} &= \int_{t \in [t_k, t_{k+1}]} \mathbf{R}_t^{b_k} (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) dt \\
 \boldsymbol{\gamma}_{b_{k+1}}^{b_k} &= \int_{t \in [t_k, t_{k+1}]} \frac{1}{2} \boldsymbol{\Omega}(\hat{\boldsymbol{\omega}}_t - \mathbf{b}_{w_t} - \mathbf{n}_w) \boldsymbol{\gamma}_t^{b_k} dt.
 \end{aligned} \tag{26}$$



离散时间预积分量中值积分

At the beginning, $\alpha_{b_k}^{b_k}$ and $\beta_{b_k}^{b_k}$ are $\mathbf{0}$, and $\gamma_{b_k}^{b_k}$ is identity quaternion. The mean of α , β , and γ in (26) is propagated step by step as follows. Note that the additive noise terms $\mathbf{n}_a$ and $\mathbf{n}_w$ are zero mean. This results in estimated values of the preintegration terms, marked by $(\hat{\cdot})$ as

$$\begin{aligned}\hat{\alpha}_{i+1}^{b_k} &= \hat{\alpha}_i^{b_k} + \hat{\beta}_i^{b_k} \delta t + \frac{1}{2} \mathbf{R}(\hat{\gamma}_i^{b_k})(\hat{\mathbf{a}}_i - \mathbf{b}_{a_i}) \delta t^2 \\ \hat{\beta}_{i+1}^{b_k} &= \hat{\beta}_i^{b_k} + \mathbf{R}(\hat{\gamma}_i^{b_k})(\hat{\mathbf{a}}_i - \mathbf{b}_{a_i}) \delta t \\ \hat{\gamma}_{i+1}^{b_k} &= \hat{\gamma}_i^{b_k} \otimes \begin{bmatrix} 1 \\ \frac{1}{2}(\hat{\omega}_i - \mathbf{b}_{w_i}) \delta t \end{bmatrix}\end{aligned}\tag{27}$$

误差卡尔曼滤波

1、什么是误差卡尔曼滤波？

2、为什么需要误差卡尔曼滤波

- The orientation error-state is minimal (*i.e.*, it has the same number of parameters as degrees of freedom), avoiding issues related to over-parametrization (or redundancy) and the consequent risk of singularity of the involved covariances matrices, resulting typically from enforcing constraints.

The error-state system is always operating close to the origin, and therefore far from possible parameter singularities, gimbal lock issues, or the like, providing a guarantee that the linearization validity holds at all times.

The error-state is always small, meaning that all second-order products are negligible. This makes the computation of Jacobians very easy and fast. Some Jacobians may even be constant or equal to available state magnitudes.

The error dynamics are slow because all the large-signal dynamics have been integrated in the nominal-state. This means that we can apply KF corrections (which are the only means to observe the errors) at a lower rate than the predictions.

3、误差卡尔曼滤波相关概念

Magnitude	True	Nominal	Error	Composition	Measured	Noise
Full state ⁽¹⁾	$\mathbf{x}_t$	$\mathbf{x}$	$\delta\mathbf{x}$	$\mathbf{x}_t = \mathbf{x} \oplus \delta\mathbf{x}$		
Position	$\mathbf{p}_t$	$\mathbf{p}$	$\delta\mathbf{p}$	$\mathbf{p}_t = \mathbf{p} + \delta\mathbf{p}$		
Velocity	$\mathbf{v}_t$	$\mathbf{v}$	$\delta\mathbf{v}$	$\mathbf{v}_t = \mathbf{v} + \delta\mathbf{v}$		
Quaternion ^(2,3)	$\mathbf{q}_t$	$\mathbf{q}$	$\delta\mathbf{q}$	$\mathbf{q}_t = \mathbf{q} \otimes \delta\mathbf{q}$		
Rotation matrix ^(2,3)	$\mathbf{R}_t$	$\mathbf{R}$	$\delta\mathbf{R}$	$\mathbf{R}_t = \mathbf{R} \delta\mathbf{R}$		
Angles vector ⁽⁴⁾			$\delta\boldsymbol{\theta}$	$\delta\mathbf{q} = e^{\delta\boldsymbol{\theta}/2}$ $\delta\mathbf{R} = e^{[\delta\boldsymbol{\theta}]_{\times}}$		
Accelerometer bias	$\mathbf{a}_{bt}$	$\mathbf{a}_b$	$\delta\mathbf{a}_b$	$\mathbf{a}_{bt} = \mathbf{a}_b + \delta\mathbf{a}_b$		$\mathbf{a}_w$
Gyrometer bias	$\boldsymbol{\omega}_{bt}$	$\boldsymbol{\omega}_b$	$\delta\boldsymbol{\omega}_b$	$\boldsymbol{\omega}_{bt} = \boldsymbol{\omega}_b + \delta\boldsymbol{\omega}_b$		$\boldsymbol{\omega}_w$
Gravity vector	$\mathbf{g}_t$	$\mathbf{g}$	$\delta\mathbf{g}$	$\mathbf{g}_t = \mathbf{g} + \delta\mathbf{g}$		
Acceleration	$\mathbf{a}_t$				$\mathbf{a}_m$	$\mathbf{a}_n$
Angular rate	$\boldsymbol{\omega}_t$				$\boldsymbol{\omega}_m$	$\boldsymbol{\omega}_n$

VINS论文中对误差卡尔曼的描述

Then, we deal with the covariance propagation. Since the four-dimensional rotation quaternion $\gamma_t^{b_k}$ is overparameterized, we define its error term as a perturbation around its mean

$$\gamma_t^{b_k} \approx \hat{\gamma}_t^{b_k} \otimes \begin{bmatrix} 1 \\ \frac{1}{2}\delta\theta_t^{b_k} \end{bmatrix} \quad (28)$$

误差状态帮助IMU预积分计算协方差时避免过参数化

连续时间IMU误差状态传递的推导

$$\begin{aligned}
 \begin{bmatrix} \delta \dot{\boldsymbol{\alpha}}_t^{b_k} \\ \delta \dot{\boldsymbol{\beta}}_t^{b_k} \\ \delta \dot{\boldsymbol{\theta}}_t^{b_k} \\ \delta \dot{\mathbf{b}}_{a_t} \\ \delta \dot{\mathbf{b}}_{w_t} \end{bmatrix} &= \begin{bmatrix} 0 & \mathbf{I} & 0 & 0 & 0 \\ 0 & 0 & -\mathbf{R}_t^{b_k} [\hat{\mathbf{a}}_t - \mathbf{b}_{a_t}] \times & -\mathbf{R}_t^{b_k} & 0 \\ 0 & 0 & -[\hat{\boldsymbol{\omega}}_t - \mathbf{b}_{w_t}] \times & 0 & -\mathbf{I} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \delta \boldsymbol{\alpha}_t^{b_k} \\ \delta \boldsymbol{\beta}_t^{b_k} \\ \delta \boldsymbol{\theta}_t^{b_k} \\ \delta \mathbf{b}_{a_t} \\ \delta \mathbf{b}_{w_t} \end{bmatrix} \\
 &+ \begin{bmatrix} 0 & 0 & 0 & 0 \\ -\mathbf{R}_t^{b_k} & 0 & 0 & 0 \\ 0 & -\mathbf{I} & 0 & 0 \\ 0 & 0 & \mathbf{I} & 0 \\ 0 & 0 & 0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{n}_a \\ \mathbf{n}_w \\ \mathbf{n}_{b_a} \\ \mathbf{n}_{b_w} \end{bmatrix} = \mathbf{F}_t \delta \mathbf{z}_t^{b_k} + \mathbf{G}_t \mathbf{n}_t.
 \end{aligned}
 \tag{29}$$

基于中值积分的离散时间预积分误差传递

$$\delta x_{k+1} = (I + F_t \delta t) \delta x_k + G_t \delta t * n_t$$

$$\begin{bmatrix} \delta\alpha_{k+1} \\ \delta\theta_{k+1} \\ \delta\beta_{k+1} \\ \delta b_{ak+1} \\ \delta b_{wk+1} \end{bmatrix} = \begin{bmatrix} I & f_{01} & \delta t & f_{03} & f_{04} \\ 0 & f_{11} & 0 & 0 & -\delta t \\ 0 & f_{21} & I & f_{23} & f_{24} \\ 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & I \end{bmatrix} \begin{bmatrix} \delta\alpha_k \\ \delta\theta_k \\ \delta\beta_k \\ \delta b_{ak} \\ \delta b_{wk} \end{bmatrix} + \begin{bmatrix} v_{00} & v_{01} & v_{02} & v_{03} & 0 & 0 \\ 0 & \frac{-\delta t}{2} & 0 & \frac{-\delta t}{2} & 0 & 0 \\ \frac{-R_k \delta t}{2} & v_{21} & \frac{-R_{k+1} \delta t}{2} & v_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & \delta t & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta t \end{bmatrix} \begin{bmatrix} n_{ak} \\ n_{wk} \\ n_{ak+1} \\ n_{wk+1} \\ n_{ba} \\ n_{bw} \end{bmatrix}$$

$$f_{01} = \frac{\delta t}{2} f_{21} = -\frac{1}{4} R_k (\hat{a}_k - b_{a_k})^\wedge \delta t^2 - \frac{1}{4} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \left[I - \left(\frac{\hat{\omega}_k + \hat{\omega}_{k+1}}{2} - b_{\omega_k} \right)^\wedge \delta t \right] \delta t^2$$

$$f_{03} = -\frac{1}{4} (R_k + R_{k+1}) \delta t^2 \quad f_{04} = \frac{\delta t}{2} f_{24} = \frac{1}{4} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \delta t^3$$

$$f_{11} = I - \left(\frac{\hat{\omega}_k + \hat{\omega}_{k+1}}{2} - b_{\omega_k} \right)^\wedge \delta t$$

$$f_{21} = -\frac{1}{2} R_k (\hat{a}_k - b_{a_k})^\wedge \delta t - \frac{1}{2} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \left[I - \left(\frac{\hat{\omega}_k + \hat{\omega}_{k+1}}{2} - b_{\omega_k} \right)^\wedge \delta t \right] \delta t$$

$$f_{23} = -\frac{1}{2} (R_k + R_{k+1}) \delta t \quad f_{24} = \frac{1}{2} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \delta t^2$$

$$v_{00} = -\frac{1}{4} R_k \delta t^2$$

$$v_{01} = v_{03} = \frac{\delta t}{2} v_{21} = \frac{1}{4} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \delta t^2 \frac{\delta t}{2}$$

$$v_{02} = -\frac{1}{4} R_{k+1} \delta t^2 \quad v_{21} = v_{23} = \frac{1}{4} R_{k+1} (\hat{a}_{k+1} - b_{a_k})^\wedge \delta t^2$$

离散时间角度误差的推导

$$\dot{\delta\theta} = -(\hat{\omega}_t - b_{\omega_t})^{\wedge} \delta\theta - n_{\omega} - \delta b_{\omega_t}$$

离散时间速度误差的推导

$$\delta \dot{\beta} = -R_t (\hat{a}_t - b_{a_t})^\wedge \delta \theta - R_t \delta b_{a_t} - R_t n_a$$

离散时间位置误差的推导

离散时间零偏误差的推导

预积分中关于imu零偏的建模

$$\begin{aligned}
 \alpha_{b_{k+1}}^{b_k} &= \iint_{t \in [t_k, t_{k+1}]} \mathbf{R}_t^{b_k} (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) dt^2 \\
 \beta_{b_{k+1}}^{b_k} &= \int_{t \in [t_k, t_{k+1}]} \mathbf{R}_t^{b_k} (\hat{\mathbf{a}}_t - \mathbf{b}_{a_t} - \mathbf{n}_a) dt \\
 \gamma_{b_{k+1}}^{b_k} &= \int_{t \in [t_k, t_{k+1}]} \frac{1}{2} \boldsymbol{\Omega} (\hat{\boldsymbol{\omega}}_t - \mathbf{b}_{w_t} - \mathbf{n}_w) \gamma_t^{b_k} dt. \quad (26)
 \end{aligned}$$

$$\alpha_{b_{k+1}}^{b_k} \approx \hat{\alpha}_{b_{k+1}}^{b_k} + \mathbf{J}_{b_a}^\alpha \delta \mathbf{b}_{a_k} + \mathbf{J}_{b_w}^\alpha \delta \mathbf{b}_{w_k}$$

$$\beta_{b_{k+1}}^{b_k} \approx \hat{\beta}_{b_{k+1}}^{b_k} + \mathbf{J}_{b_a}^\beta \delta \mathbf{b}_{a_k} + \mathbf{J}_{b_w}^\beta \delta \mathbf{b}_{w_k}$$

$$\gamma_{b_{k+1}}^{b_k} \approx \hat{\gamma}_{b_{k+1}}^{b_k} \otimes \left[\frac{1}{2} \mathbf{J}_{b_w}^\gamma \delta \mathbf{b}_{w_k} \right]$$

如何计算预积分量关于零偏的雅克比

$$\mathbf{J}_{t+\delta t} = (\mathbf{I} + \mathbf{F}_t \delta t) \mathbf{J}_t, \quad t \in [k, k+1].$$

预积分协方差矩阵的传递

代码对照讲解